

(retired) Raja Menon.

ATV, with its suitably muted acronym, was a euphemism for a long-delayed project. Shrouded in obsessive secrecy for decades, it has been under the direct supervision of the prime minister, who also chairs ATV's apex committee. Nearly 200 naval officers and technicians are directly involved in the project that is managed by a vice-admiral who functions out of ATV headquarters in Delhi Cantonment. Funding was never a problem, even during the lean days of defence spending, like in the pre-1990s. An estimated Rs 2,000 crore was spent even before work on the submarine was started. The excessive secrecy, say experts, was based on a misinterpretation of the Non-Proliferation Treaty (NPT)—that building a nuclear submarine would be a violation. There was, therefore, a lack of accountability, which harmed the project.

Project officials in Vizag are now sealing the reactor with a special shield and plugging in the control systems, turbines and piping. The next few months are critical. After the reactor compartment is sealed, the tail sector—which includes the propeller and the shaft—will be welded in and the submarine will be ready. By April next year, the dry dock will be flooded and the vessel will be officially launched.

After it hits the water, the nuclear reactor will be jump-started and the submarine's propellers—seven highly-skewed brass blades—will be tested. After the reactor and all its associated control systems are successfully proven, the

THE NUCLEAR TRIAD

Nuclear weapons can be launched from three types of platforms. ATV forms the elusive sea-based third leg of the deterrent.

AIR-LAUNCHED WEAPONS  Nuclear weapons are dropped from bombers like Mirage-2000 aircraft of IAF. VULNERABILITY: Can carry only tactical N-weapons. Needs dedicated bombers for larger N-weapons. Can be shot down or intercepted.	LAND-BASED SYSTEMS Nuclear-tipped ballistic missiles are mounted on trucks and trains. The Agni-I and II are rail mobile systems, Agni-III is on a special 16-wheeled cross-country vehicle. VULNERABILITY: Vehicles and trains can be tracked and intercepted. Heavily dependent on rail/road infrastructure. Vulnerable to destruction of bridges.
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"It is a strategic project and things are developing according to the schedule."

A.K. ANTONY, DEFENCE MINISTER

submarine will be towed out of the harbour for extensive sea trials lasting over a year before it is inducted into the navy around 2010. While the impending launch of ATV is reason for cheer, the actual fielding of a secure second-strike capability is still three years away. This is the time it will take to integrate and successfully test fire the missile from the submarine.

Without its nuclear missiles, the submarine is just a platform. The missile is being concurrently developed under an equally-classified programme. Announcing its

successful test in April last year, DRDO chief M. Natarajan called it "a strategic system which I cannot talk about".

The enigmatic two-stage missile—dubbed K-15 under the Defence Research and Development Organisation's (DRDO) Sagarika (oceanic) project—is a technological breakthrough. Rapidly ejected from the submarine's launcher by igniting an underwater gas booster, it rises nearly 5 km above the ocean. When it reaches a pre-determined height, it ignites a solid booster and travels to a range of nearly 750 km. Tested three times from a specially-designed submersible pontoon, the yet-to-be-named "naval missile" is another feather in India's cap.

The 100-member crew, which will

ATV PROJECT SITES

Like pieces of a complex jigsaw puzzle, ATV project sites are scattered across the country.

KALPAKKAM

Indira Gandhi Atomic Research Centre near Chennai fabricates ATV's light water nuclear reactor.

VISAKHAPATNAM:

■ ATV production line at the Ship Building Centre.

■ School of Advanced Underwater Warfare (SAUV) for training ATV crews.

■ Indigenously developed control room simulator.

■ Evolution of strategic submarine operational doctrines.

RAMBILLI

Strategic submarine base south of Vizag will be commissioned by 2011.

DELHI

ATV project headquarters.

RATNAHALLI

Rare Materials Project near Mysore supplies enriched uranium for ATV reactor.

HYDERABAD

Sagarika complex is fabricating and developing SLBM. Project began in 1994.

BALASORE

Special underwater launch test platform for test-firing ATV's missile.

KOCHI

Naval Physical and Oceanographic Laboratory developing ATV sensors.